

Coding & Solution

Date	30 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented code and the overall solution. The submission demonstrates working functionality, clean code practices, and alignment with the defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/hemanth5267/AI-ML-Track-Project-Template
Programming Language(s)	Python 3.10+, HTML5, CSS3
Framework(s) Used	Flask (Backend Web Framework), Bootstrap 5 (Frontend UI)
Key Features Implemented	• Crop recommendation using ML (Logistic Regression, K-Means Clustering) • Data preprocessing: null handling, outlier treatment, feature scaling • Univariate, Bivariate & Multivariate analysis with Matplotlib & Seaborn • Flask web application with HTML input form and prediction result page • ML model saved as .pkl file using Pickle for fast reuse • Dataset: Crop_recommendation.csv (2200 records, 8 features) from Kaggle
Pending / Incomplete Features	• Cloud deployment (AWS/Heroku) — planned for future release • Weather API integration for live climate data • Multi-language support for regional farmers
Setup / Run Instructions	1. Clone repo: git clone https://github.com/hemanth5267/AI-ML-Track-Project-Template 2. Install dependencies: pip install flask scikit-learn pandas numpy matplotlib seaborn 3. Run Jupyter notebook: jupyter notebook OptiCrop.ipynb 4. Run Flask app: python app.py 5. Open browser: http://127.0.0.1:5000

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes

3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The OptiCrop project is developed as a solo effort by P Hemanth (Team Lead) due to unresponsive teammates. All 5 Epics including data collection, preprocessing, ML model building, and Flask application development are completed individually. The Jupyter notebook (OptiCrop.ipynb) contains all analysis and ML code. The GitHub repository contains the complete project structure following the 8-folder template provided by SmartBridge Skill Wallet.